

Module 8 -- Grad Cafe: Analytical Summary

This report summarizes the cleaned Grad Cafe dataset and the SciPy statistical analyses run in module_8.ipynb. All figures are generated by code from the cleaned DataFrame; the text interprets them.

Cleaned dataset: 29,141 rows x 37 columns (from 30,000 raw rows).

Q1. Average GRE of accepted PhD applicants

-> 261.81 (n = 534 applicants with a valid GRE).

Q2. Average GPA of accepted Master's applicants

-> 3.719 (n = 3842 applicants with a valid GPA).

Both averages use only valid, non-missing scores, so they describe the reporting subset rather than every accepted applicant.

Q3. Shortcomings of inferring decision_date

decision_date is inferred from a free-text status/notification field plus the term year. Key shortcomings:

1. Ambiguous / missing year. The notification text carries only a month and day (e.g. 'Feb 10'); we borrow the year from the term (e.g. Fall 2026). Decisions that actually land in the calendar year before or after the term are dated incorrectly, which can even make days_to_decision negative.
2. Free-text fragility. Users type statuses inconsistently ('Interview' vs 'Interviewed', extra words, typos, blanks). Any format we did not anticipate is silently dropped to a missing date.
3. No timezone / self-reported lag. Dates are entered by applicants after the fact, so they approximate rather than record the true decision date; day-level precision is optimistic.

Q4-Q6. Statistical analyses

Q4. Correlation between GPA and GRE-related measures.

Pearson GRE vs GRE V: $r = 0.284$, $p = 1.06e-36$ ($n = 1900$).

GPA vs GRE (Pearson): $r = -0.001$, $p = 0.954$ ($n = 2098$).

GRE and GRE V are positively correlated ($r=0.28$), a relationship that is statistically significant. GPA and GRE show a weak negative association (not statistically significant)

Q5. Group comparison of GPA (Accepted vs Rejected).

Test: Welch two-sample t-test (`ttest_ind, equal_var=False`); statistic = -4.831, $p = 1.37e-06$.

Accepted mean GPA = 3.763 ($n = 7102$); Rejected mean GPA = 3.782 ($n = 7986$).

Accepted applicants average 3.76 GPA vs 3.78 for rejected; the difference is statistically significant ($p=1.37e-06$), so accepted and rejected applicants do differ in reported GPA.

Q6. Chi-square: Degree vs US/International.

$\chi^2 = 772.681$, $p = 6.35e-166$, $\text{dof} = 4$.

The association between Degree and US/International is statistically significant ($p=6.35e-166$); Degree type and applicant origin are not independent.

Q7. Remaining data-quality issues

Even after cleaning, several issues remain:

- * Sparse GRE reporting. Only a small minority of rows carry any GRE score, so every GRE statistic describes a self-selected reporting subset.
- * Residual out-of-range values. Some GPAs sit outside 0-4.0 (e.g. 10-point or percentage scales); `has_valid_gpa` flags but does not repair them.
- * Self-reported, unverified entries. Programs, universities, and outcomes are free text; near-duplicate spellings survive standardization.
- * Inferred, sometimes-negative `days_to_decision` from the date logic in Q3.
- * Class imbalance. Terms and outcomes are dominated by a single value, which limits how far group comparisons generalize.

Takeaway: the dataset is now analysis-ready for descriptive EDA, but the numbers should be read as patterns in self-reported data, not ground truth.